

# Intracranial Activity of Trastuzumab Deruxtecan in HER2-Low Breast Cancer With Brain Metastases: A Systematic Review and Meta-Analysis

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## Purpose

To evaluate the intracranial efficacy of trastuzumab deruxtecan (T-DXd) in patients with HER2-low metastatic breast cancer and brain metastases, with a primary focus on intracranial objective response rate (IC-ORR).

## Background

HER2-low breast cancer, defined as HER2 immunohistochemistry (IHC) 1+ or IHC 2+/*in situ* hybridization-negative, represents a newly targetable subtype encompassing nearly half of all breast cancers. Brain metastases are a frequent and clinically significant complication of metastatic breast cancer, yet patients with active central nervous system (CNS) disease are often underrepresented in clinical trials. Trastuzumab deruxtecan (T-DXd), an antibody-drug conjugate with a cleavable linker and topoisomerase I inhibitor payload, has demonstrated substantial intracranial activity in HER2-positive disease. However, its efficacy in HER2-low disease with CNS involvement remains incompletely characterized, necessitating a systematic evaluation of available data.

## Methods

A systematic literature search was conducted across PubMed/MEDLINE, Embase, CENTRAL, ClinicalTrials.gov, and Web of Science from database inception through January 1, 2025. Eligible studies included randomized trials and prospective or retrospective clinical studies evaluating T-DXd in patients with HER2-low metastatic breast cancer and measurable brain metastases. Studies were required to report intracranial objective response rate (IC-ORR), defined as complete or partial response among CNS-evaluable patients using study-defined criteria (RECIST 1.1 or RANO-BM). A random-effects proportional meta-analysis using the Freeman-Tukey double arcsine transformation was performed to estimate pooled IC-ORR. Heterogeneity was assessed using the  $I^2$  statistic.

## Results

Five studies comprising 62 patients with HER2-low metastatic breast cancer and brain metastases were included. The pooled IC-ORR was 46% (95% CI, 27-65%), indicating clinically meaningful intracranial activity. Between-study heterogeneity was moderate ( $I^2 = 47.7\%$ ,  $p = 0.11$ ), with consistent directionality of effect across included cohorts despite variations in study design and patient populations.

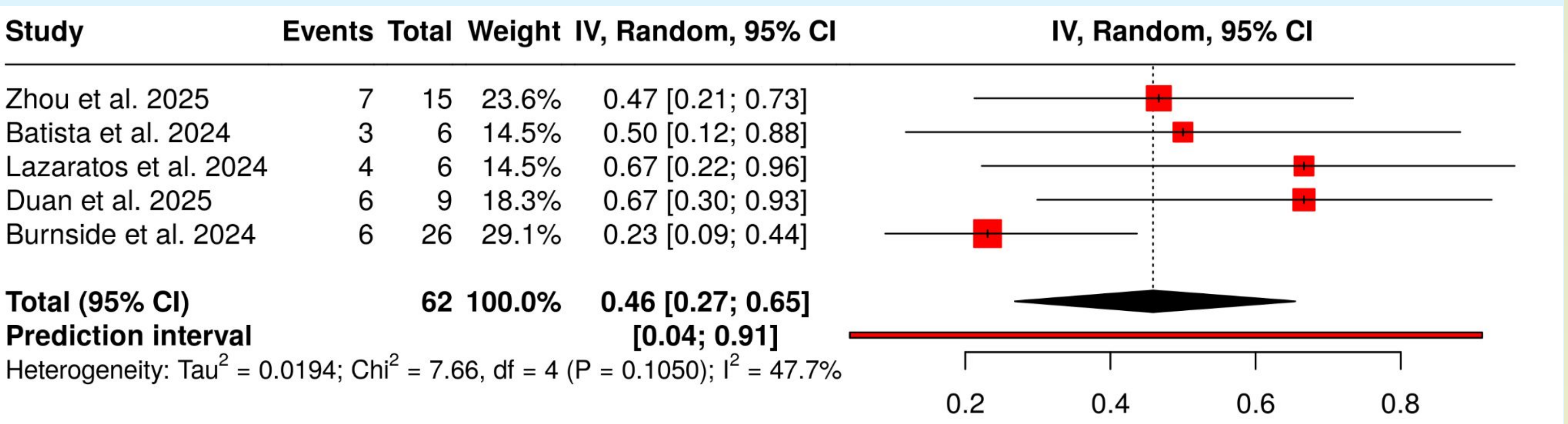


Figure 2 Forest Plot of Intracranial Objective Response Rate (IC-ORR)

| Study                            | Year | CNS-Evaluable Patients (N) | Intracranial Responders (CR+PR) | IC-ORR % (95% CI) |
|----------------------------------|------|----------------------------|---------------------------------|-------------------|
| Zhou et al.                      | 2025 | 15                         | 7                               | 46.7 (21.0–73.0)  |
| Batista et al.                   | 2024 | 6                          | 3                               | 50.0 (12.0–88.0)  |
| Lazaratos et al.                 | 2024 | 6                          | 4                               | 66.7 (22.0–96.0)  |
| Duan et al.                      | 2025 | 9                          | 6                               | 66.7 (30.0–93.0)  |
| Burnside et al.                  | 2024 | 26                         | 6                               | 23.1 (9.0–44.0)   |
| Pooled estimate (random effects) | —    | 62                         | 26                              | 46.0 (27.0–65.0)* |

Table 1 Summary of Intracranial Response Outcomes

## Implications

Trastuzumab deruxtecan demonstrates meaningful intracranial activity in HER2-low metastatic breast cancer, providing one of the first quantitative estimates of CNS response in this emerging subtype. These findings support further prospective evaluation in patients with active brain metastases and underscore the need for standardized intracranial endpoints. Incorporation of CNS-specific outcomes in future trials will be critical to refining patient selection and optimizing therapeutic strategies in HER2-low disease.